

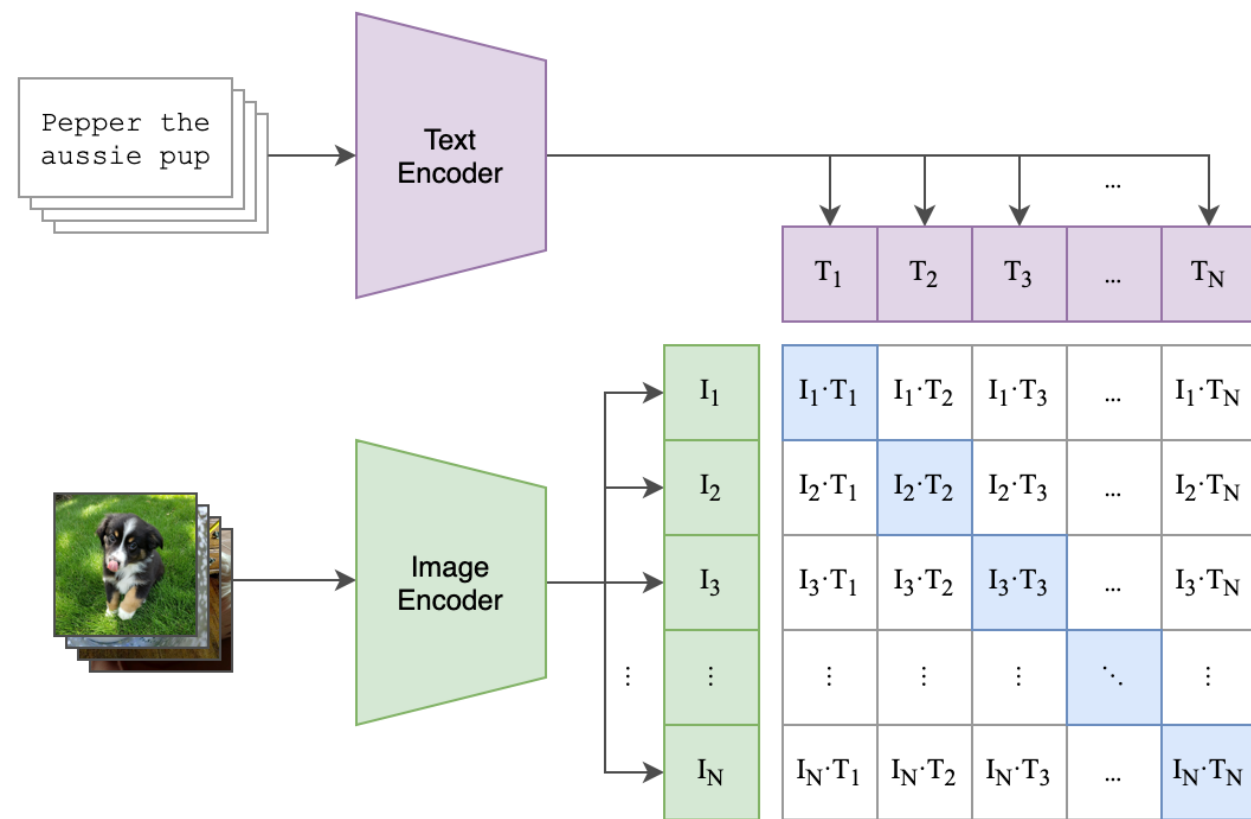
Нейросетевые решения для CV
Vision Large Models (VLM)
Диффузионные модели

Нейросетевые решения для CV

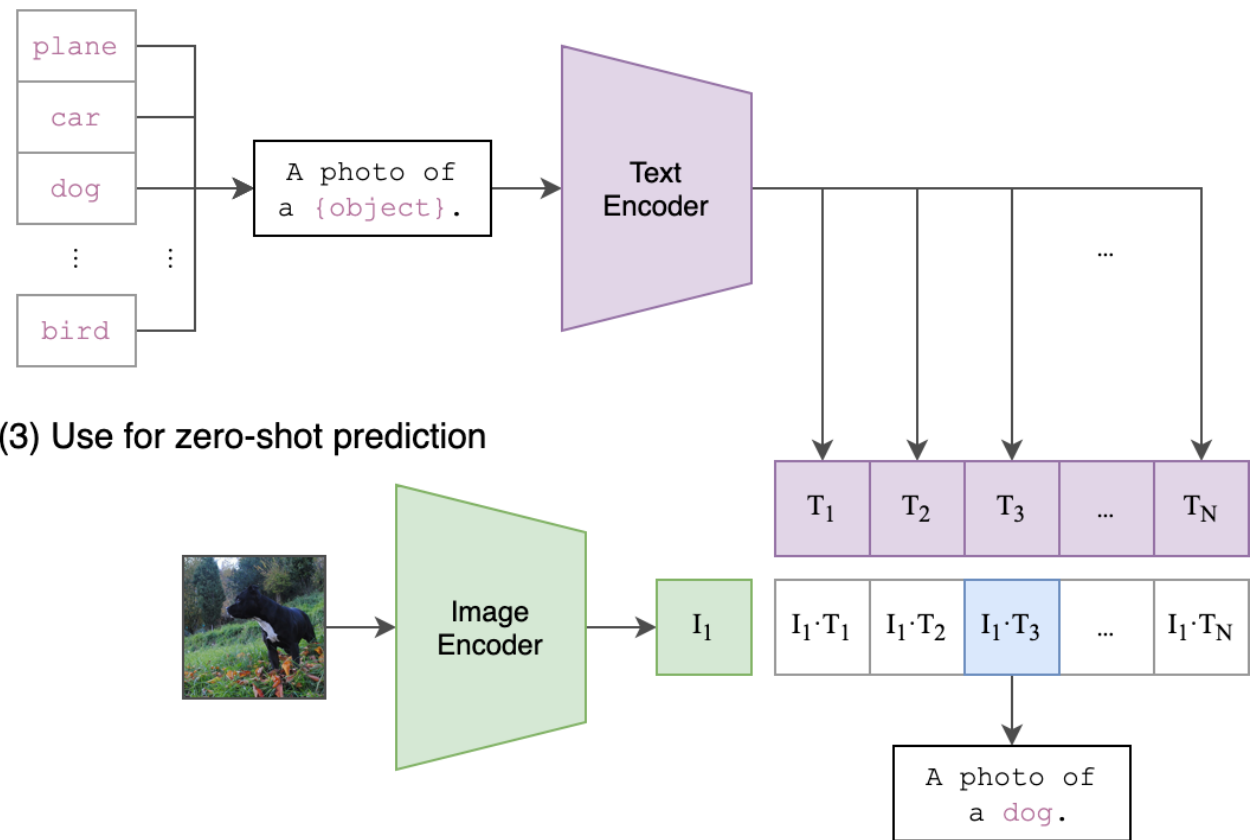
Zero-shot классификация

CLIP

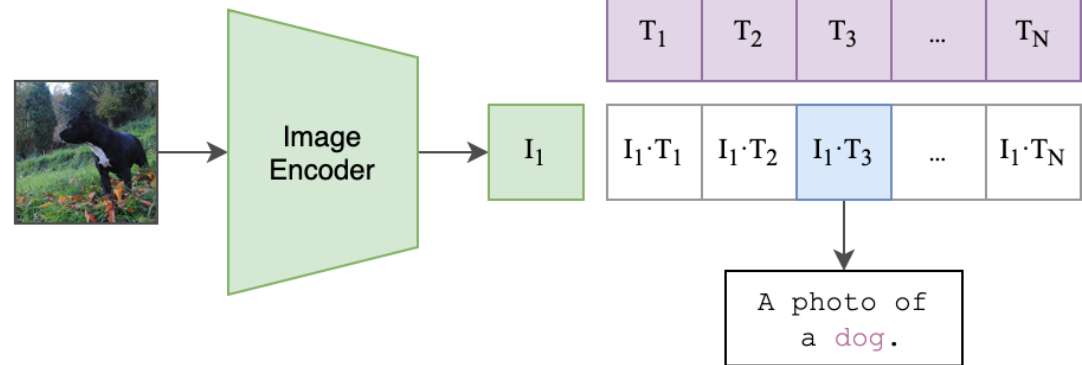
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



Zero-shot детекция

Grounding DINO

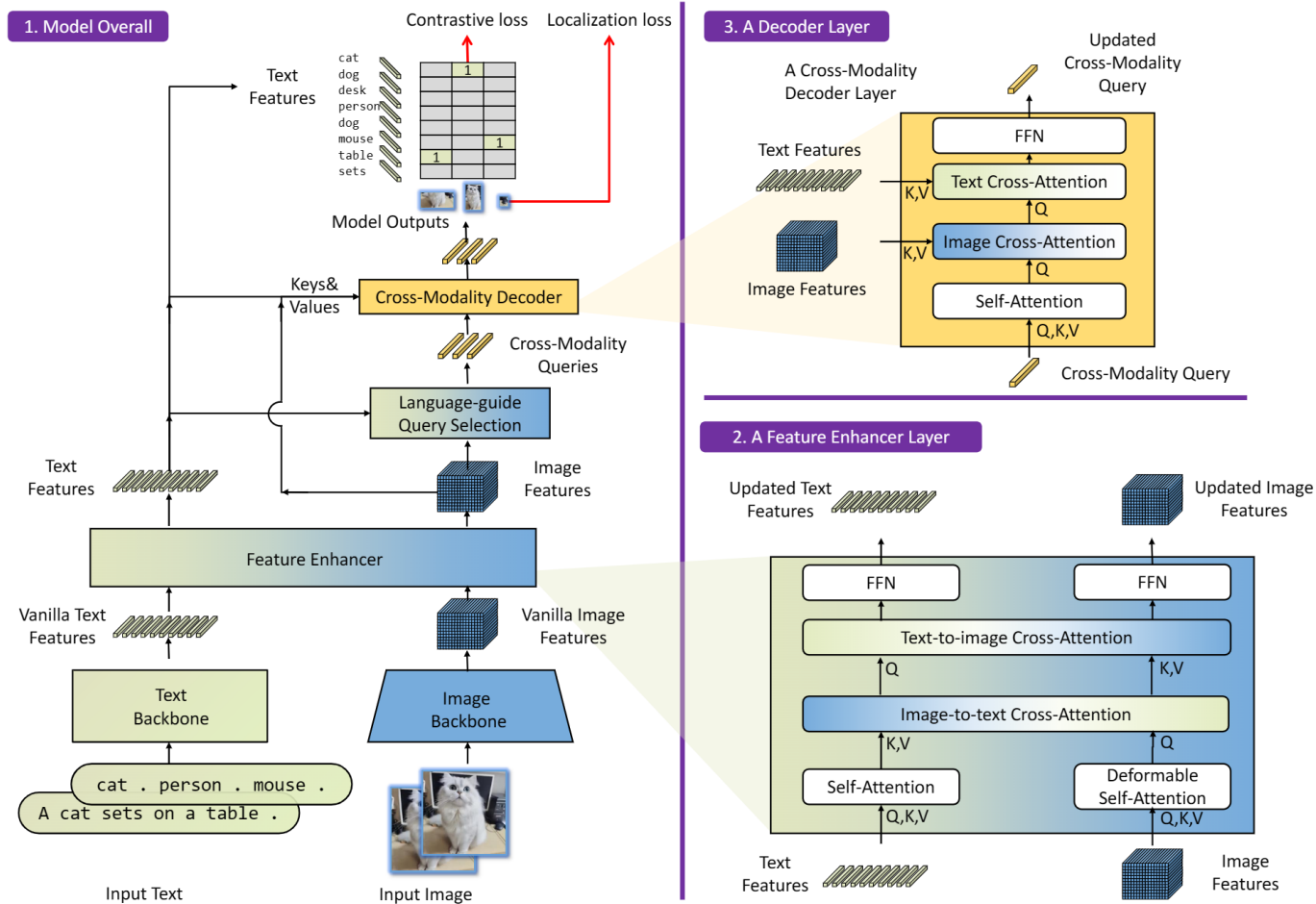
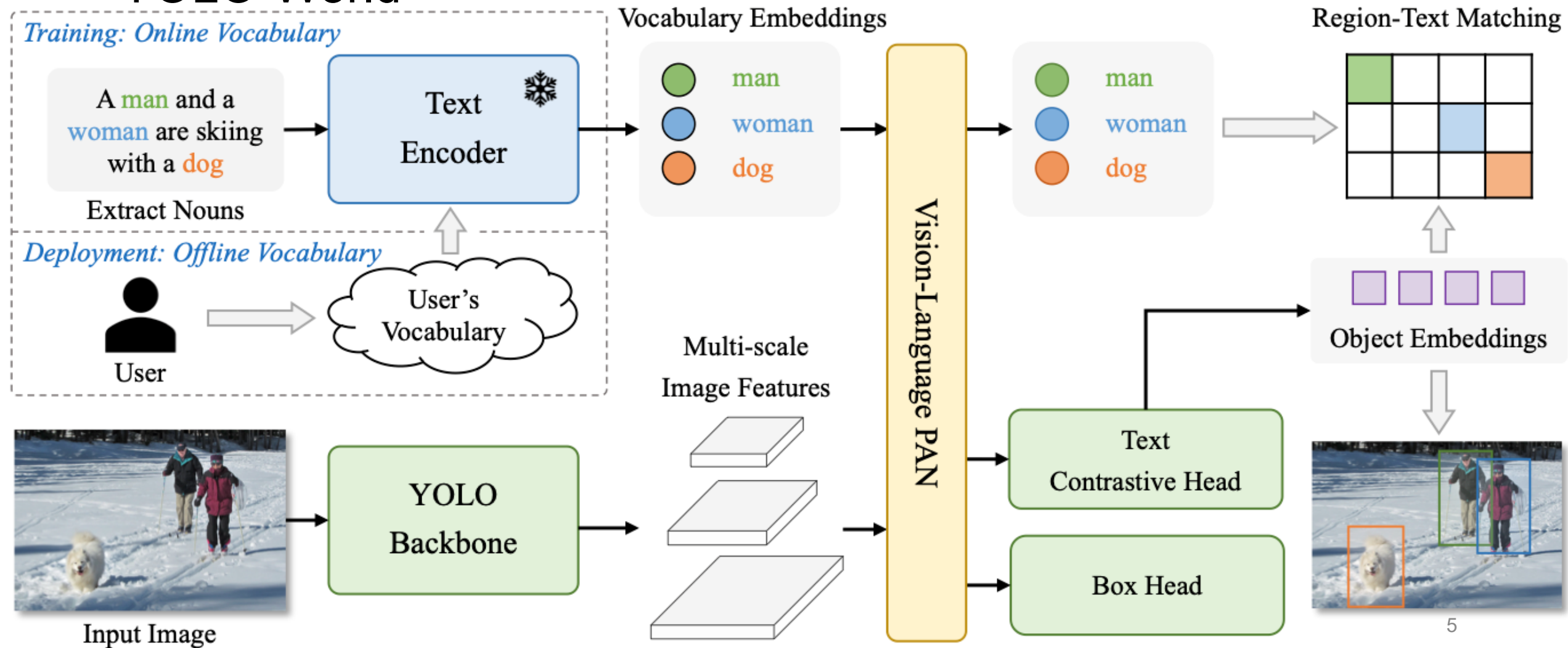


Figure 3. The framework of Grounding DINO. We present the overall framework, a feature enhancer layer, and a decoder layer in block 1, block 2, and block 3, respectively.

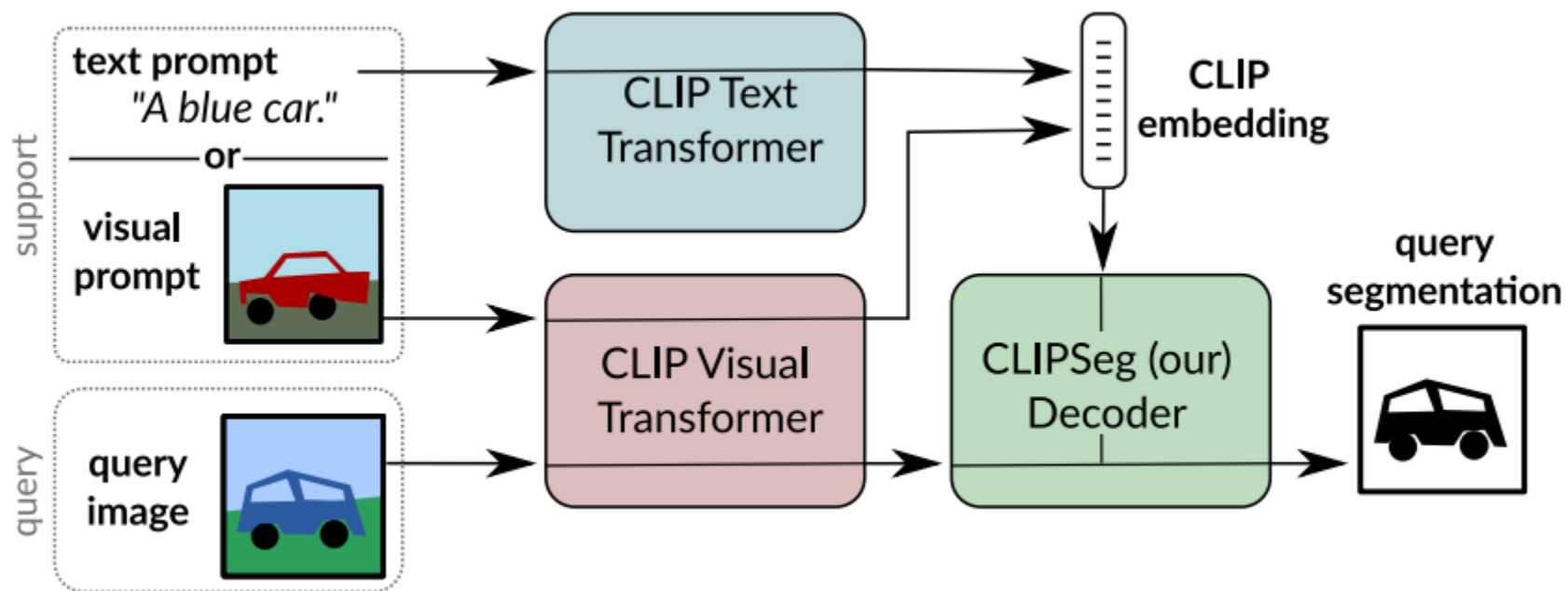
Zero-shot детекция

YOLO-World



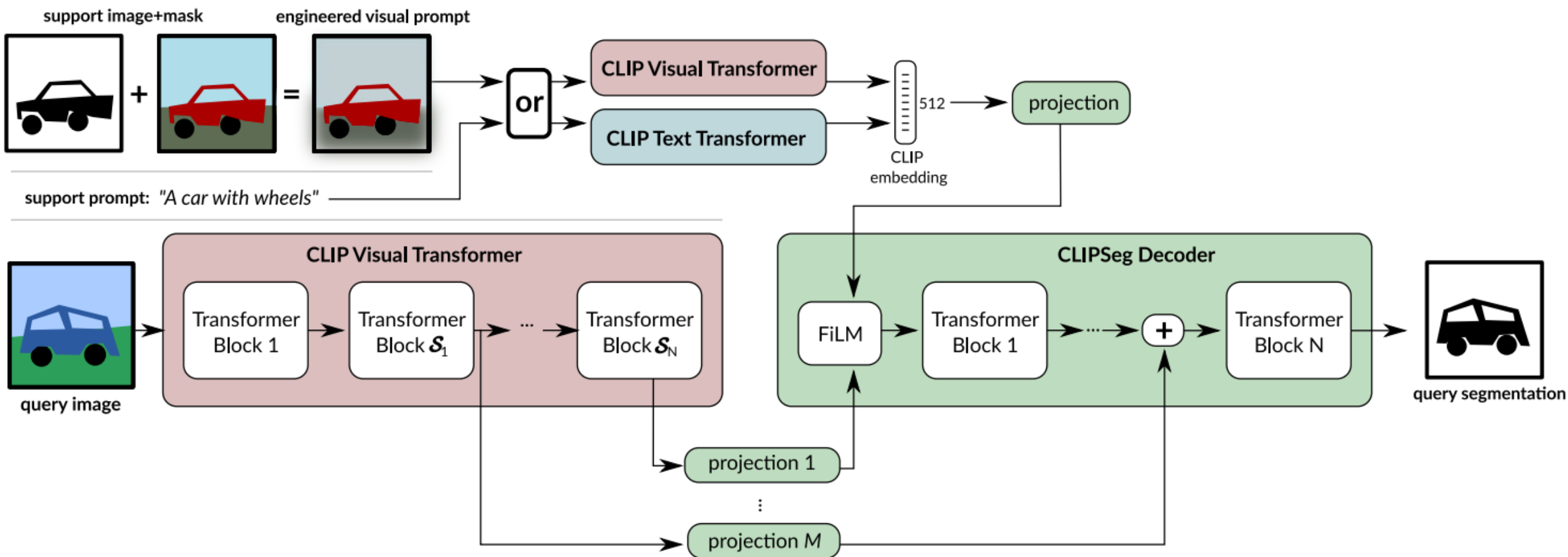
Zero-shot сегментация

CLIPSeg



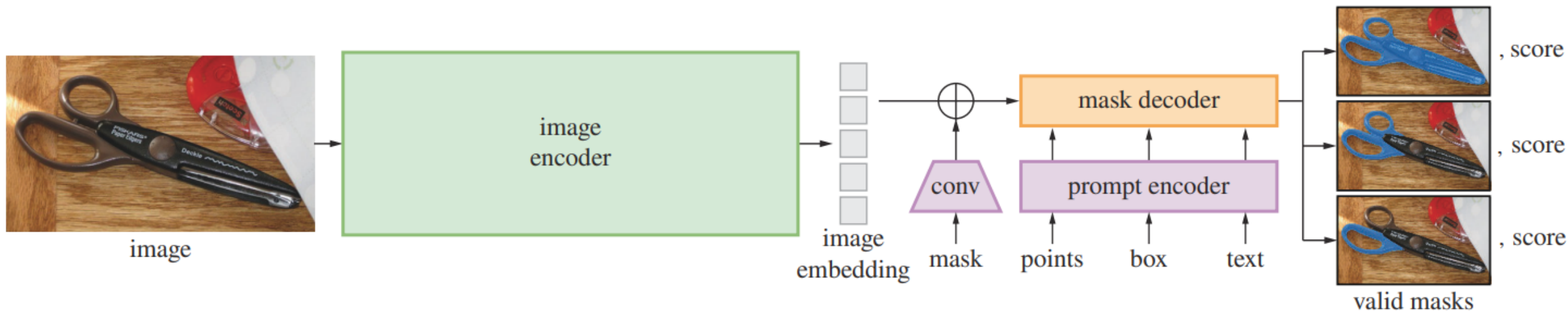
Zero-shot сегментация

CLIPSeg



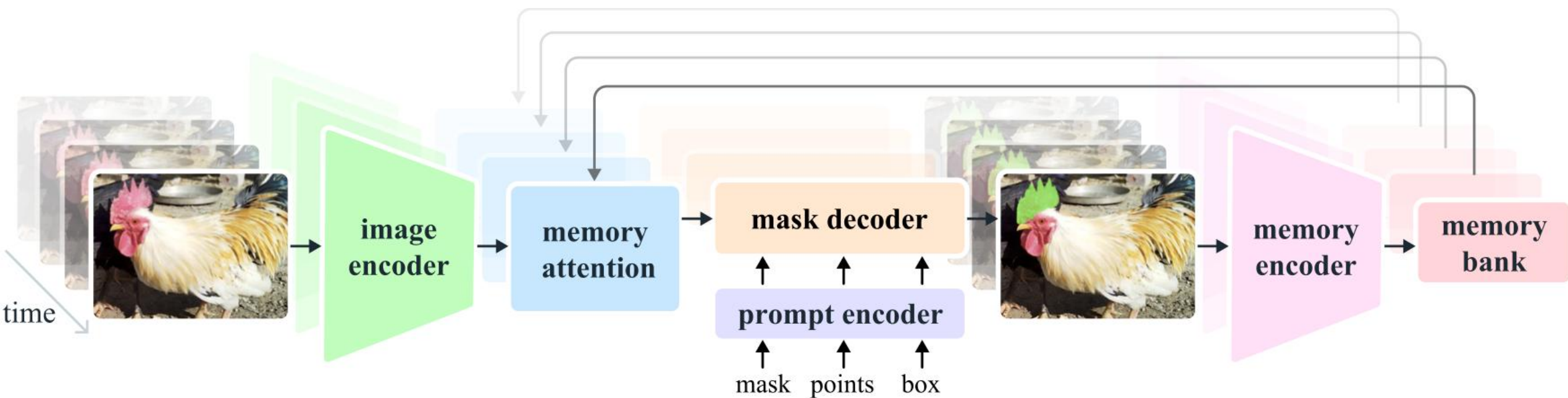
Интерактивная сегментация

Segment Anything Model (SAM)



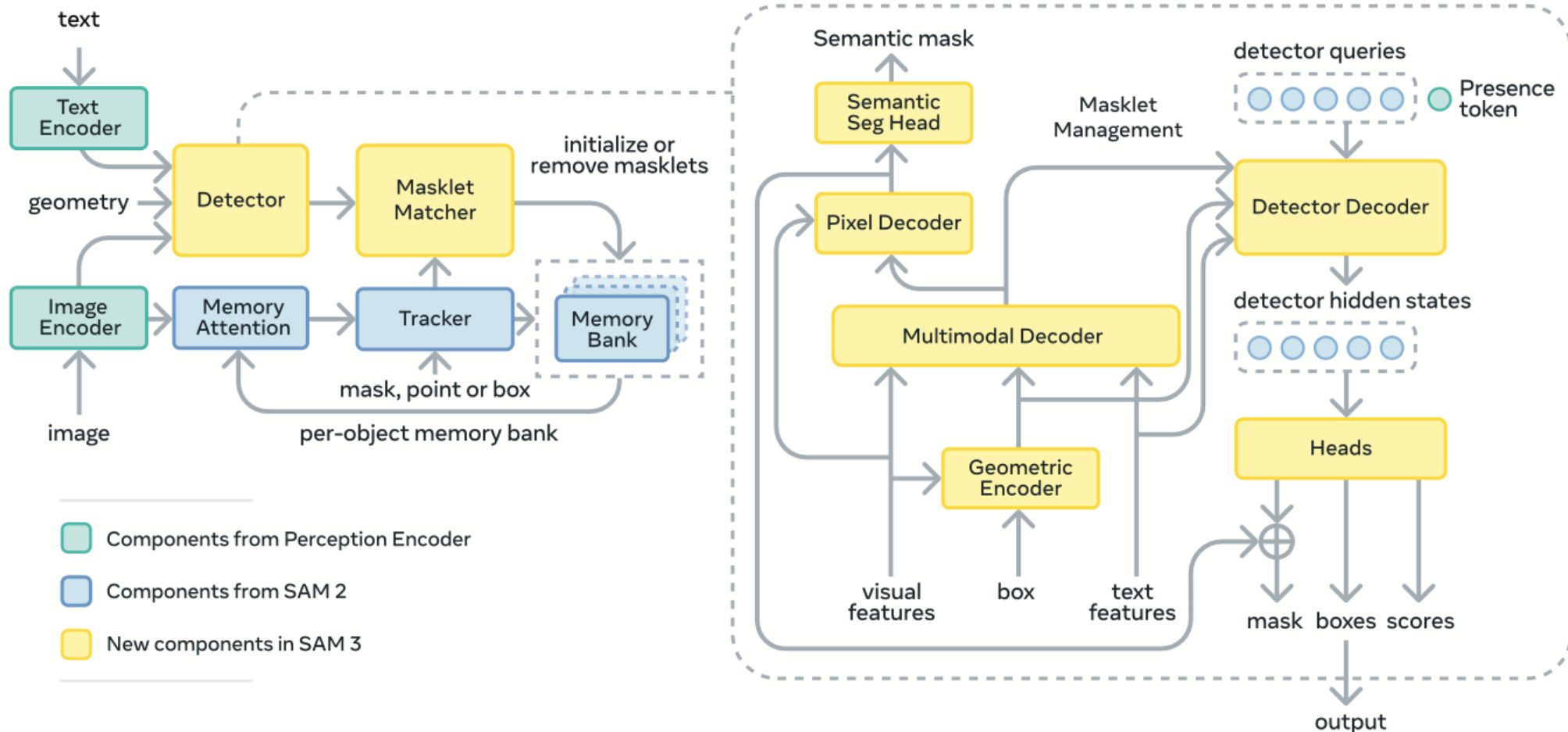
Интерактивная сегментация

Segment Anything Model 2 (SAM2)



Zero-shot instance сегментация

Segment Anything Model 3 (SAM3)



Zero-shot instance сегментация

Segment Anything Model 3 (SAM3)

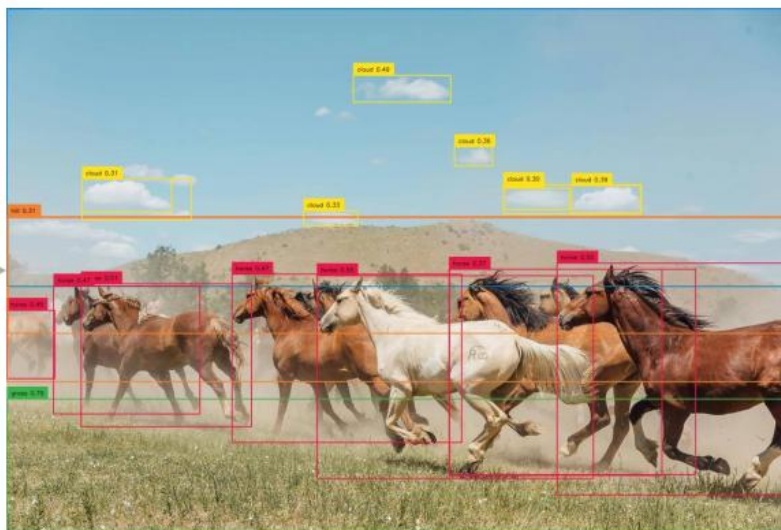
Feature	SAM 1	SAM 2	SAM 3
Zero-Shot Segmentation Quality	★★★★☆	★★★★☆	★★★★★
Small Object Performance	★★★★☆	★★★★☆	★★★★☆
Boundary Accuracy	★★★★☆	★★★★☆	★★★★★
Prompt Types Supported	Points, Boxes	Points, Boxes, Masks	Points, Boxes, Masks, Text, Examples
Video Segmentation	—	Limited	Full video support
3D / Volumetric Segmentation	—	Experimental	Supported
Inference Speed	Medium	Medium	Faster + more efficient
Memory Efficiency	High usage	Moderate usage	Optimized / lower
Fine-Tuning Support	None	Limited	Stable + recommended
Cross-Domain Robustness	Weak	Moderate	Strong
Model Size Options	Multiple sizes	Multiple sizes	Singular, large model
License Clarity	Custom	Custom	Custom

Zero-shot instance сегментация

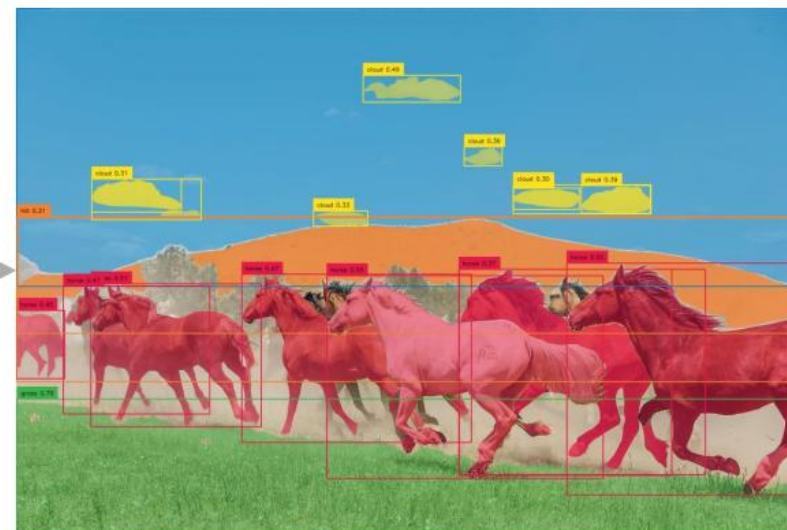
Grounded Segment Anything (GroundingDINO + SAM)



Text Prompt:
“Horse. Clouds. Grasses. Sky. Hill.”



Grounding DINO:
Detect Everything

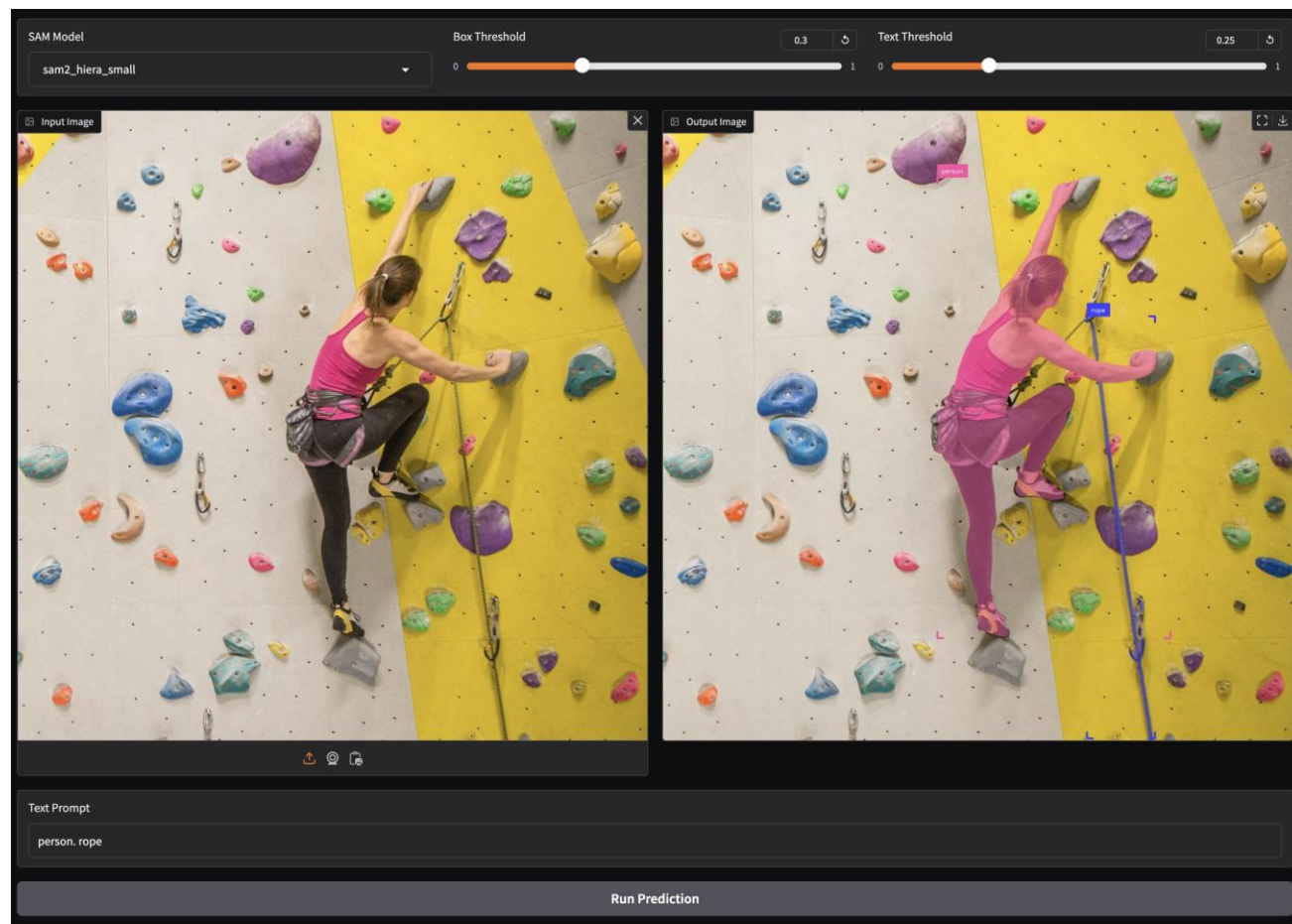


Grounded-SAM:
Detect and Segment Everything

<https://github.com/IDEA-Research/Grounded-Segment-Anything>

Zero-shot instance сегментация

LangSAM (GroundingDINO + SAM)



<https://github.com/luca-medeiros/lang-segment-anything>

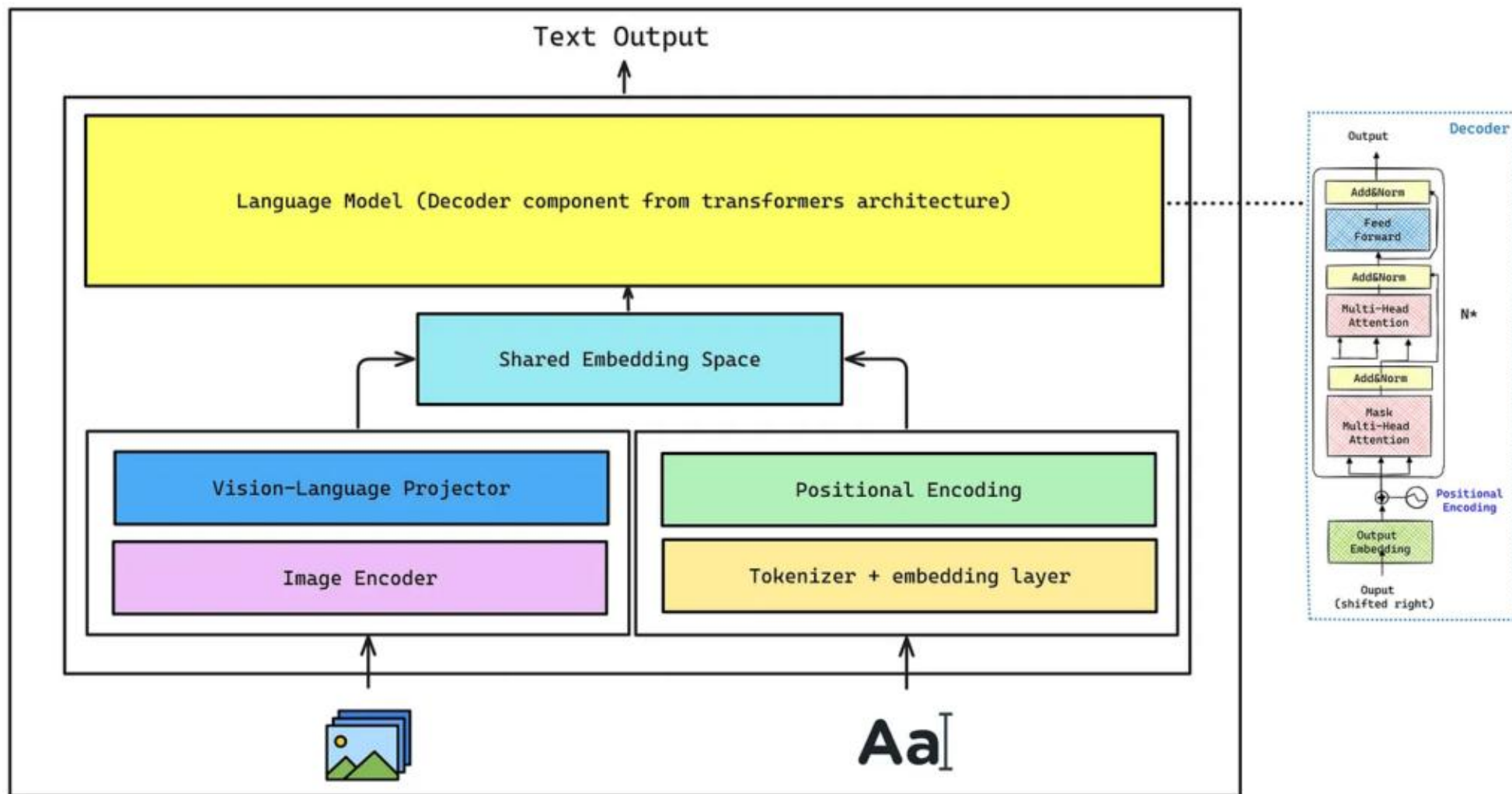
Ссылки на colab

- [YOLO-World](#)
- [CLIPSeg](#)
- [SAM2](#)
- [SAM3](#)

Vision Large Models (VLM)

VLM

Ключевые компоненты



VLM

Эволюция VLM

- **Pioneering Models (e.g., CLIP):** OpenAI's **CLIP (Contrastive Language-Image Pre-training)**, released in 2021, was a major breakthrough. CLIP was trained on a massive dataset of image-text pairs from the internet. It learned to determine how well a given text description matched an image. While it couldn't generate descriptions itself, it created a powerful shared embedding space that became a foundational block for future models.
- **Generative Models (e.g., Flamingo, LLaVA):** The next step was to move from matching to generating. Models like Google's **Flamingo** and, more recently, **LLaVA (Large Language and Vision Assistant)** were built upon these ideas. LLaVA, for instance, connects the vision backbone of CLIP with a powerful LLM (like Vicuna). By using a simple fusion mechanism and fine-tuning the model on instruction-following datasets, LLaVA demonstrated the ability to have complex, conversational dialogues about images, marking a significant step towards the VLM assistants we see today.

VLM

Применения VLM

- **Image Captioning:** Automatically generating accurate and descriptive captions for images, which is invaluable for accessibility (e.g., screen readers for the visually impaired) and content organization.



Man wearing blue mask



Colorful buttons



Hot air balloon in
foggy mountains

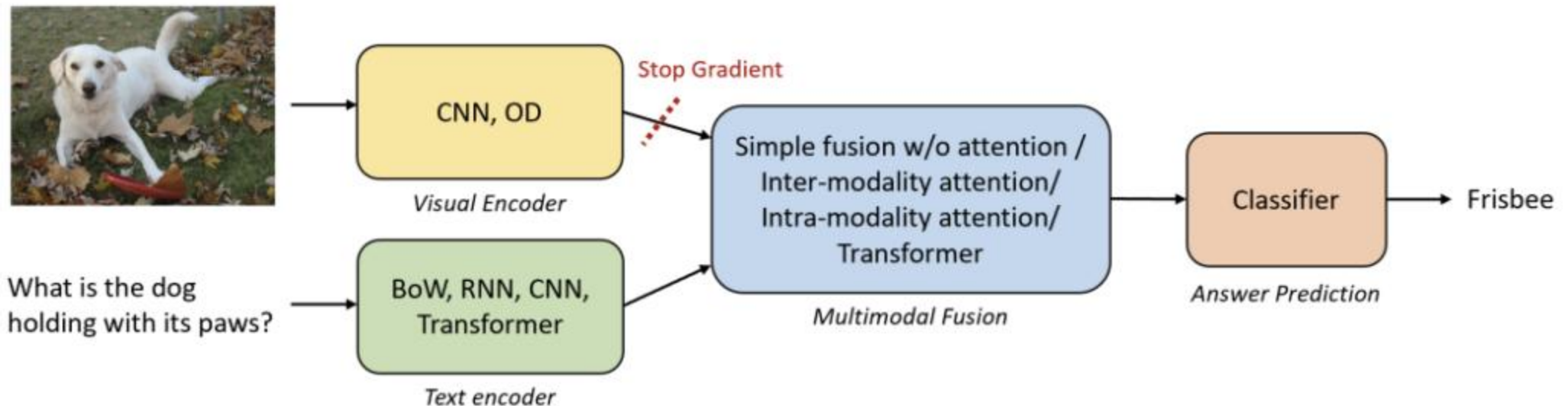


Peach fruit on a tree

VLM

Применения VLM

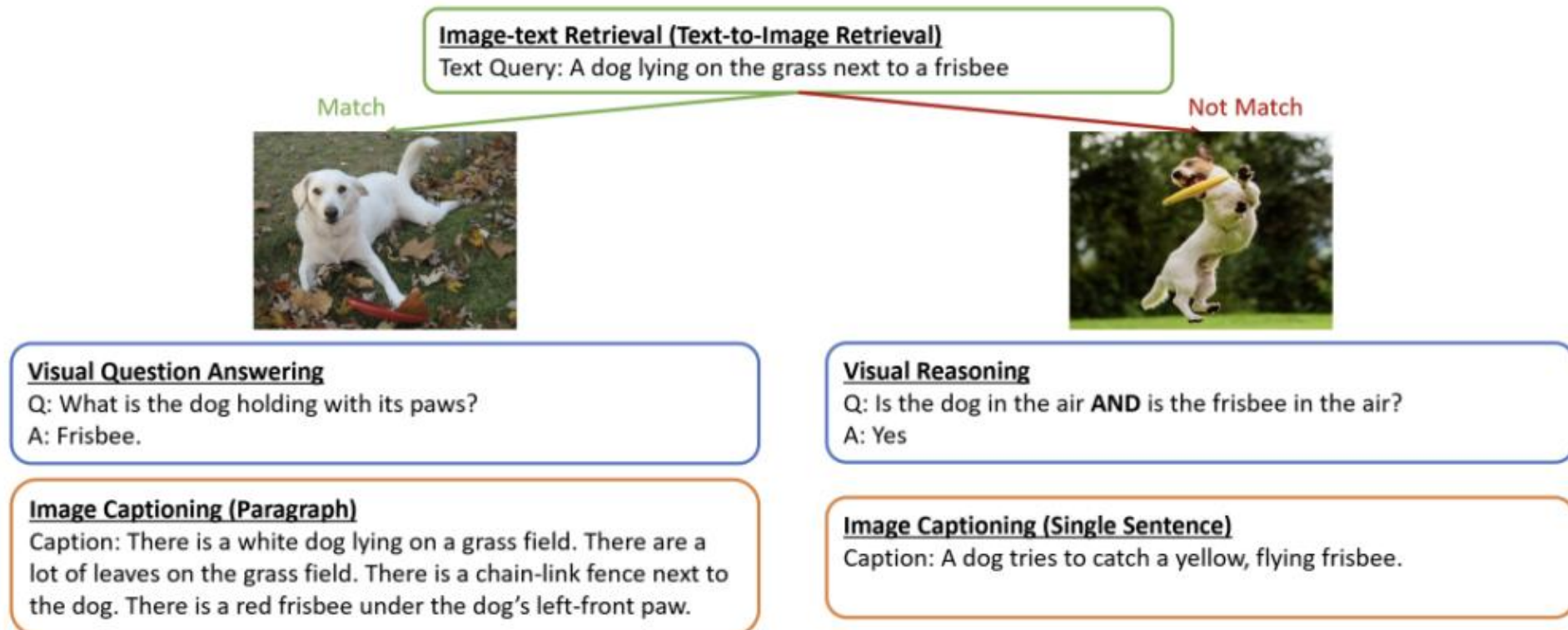
- **Visual Question Answering (VQA):** Asking specific questions about an image and getting detailed answers. For example, “How many people are sitting at the table in this cafe?” or “What color is the car on the right?”



VLM

Применения VLM

- **Image-Text Retrieval:** Revolutionizing search engines. Instead of just searching with text, you can use an image and text to find what you're looking for. For instance, uploading a picture of a chair and asking, "Where can I buy a similar dining chair but in blue?"



VLM

Дообучение VLM

[Vision Fine-Tuning](#)



unisloth

Parameter Efficient fine-tuning (PEFT)

- LoRA
- QLoRA

LoRA

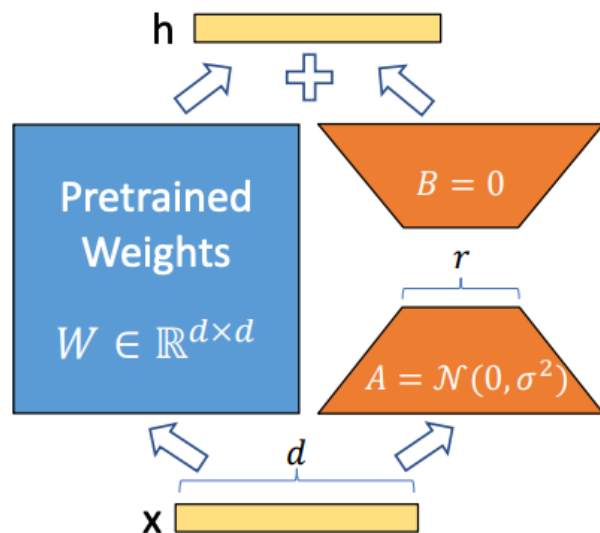


Figure 1: Our reparametrization. We only train A and B .

Линейный слой

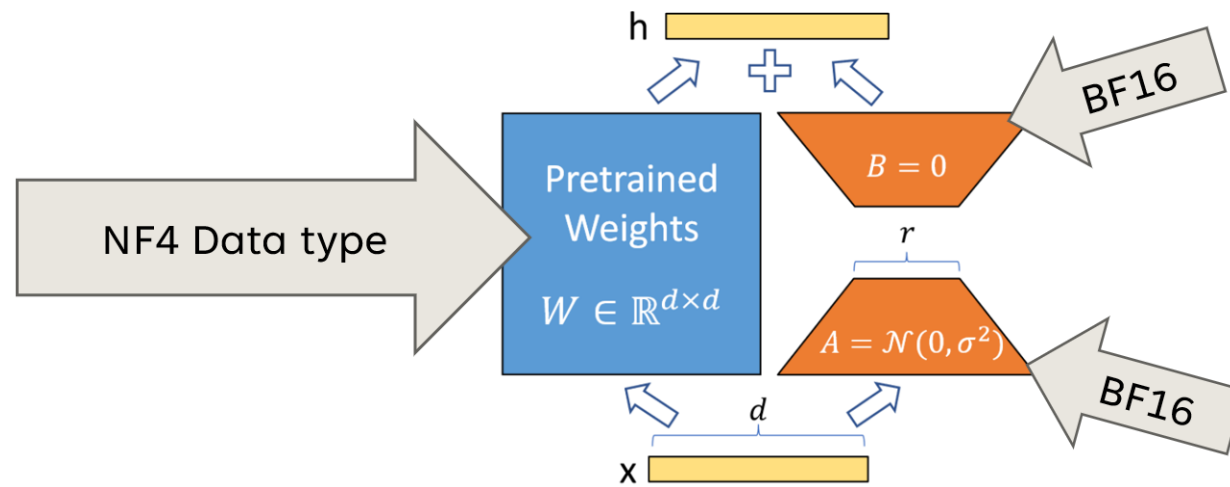
$$h = Wx$$

Линейный слой после добавления обучаемых матриц A и B

$$h = Wx + BAx$$

При этом матрица W имеет размерность $[d, d]$, а матрицы A и B — размерности $[d, r]$ и $[r, d]$ соответственно, где $r \ll d$ (обычные значения d — это 8, 16, 32). Таким образом, мы обучаем только $2 * r * d$ параметров вместо $d * d$

QLoRA



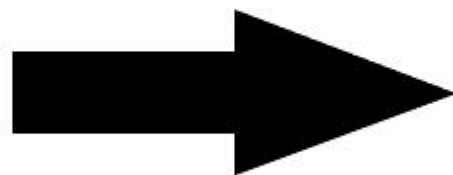
Диффузионные модели

Диффузионные модели

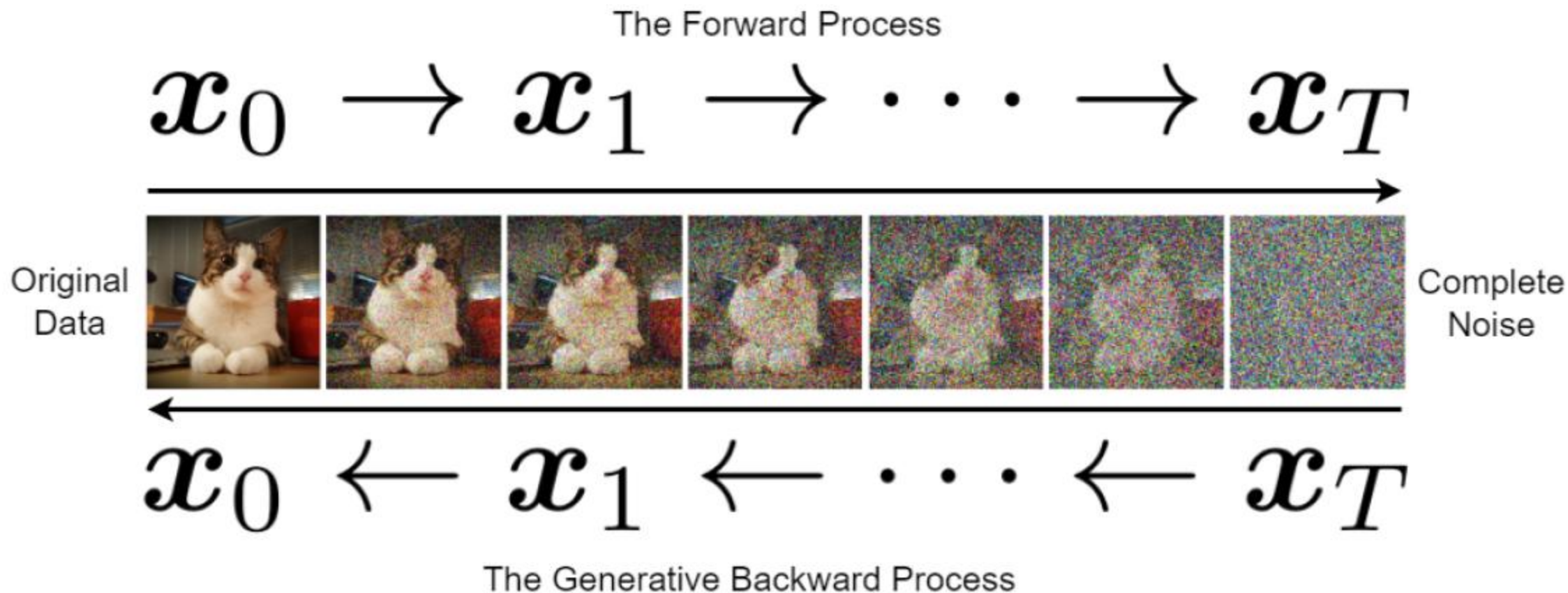
<https://colab.research.google.com/drive/10YxVSMICltPUkGD6R-Wj4EqTj0IbXRQn?usp=sharing>

<https://colab.research.google.com/drive/1xK5CALh7frhbKLg5lCBAiHCSA-AvVfw4?usp=sharing>

Диффузионные модели



Диффузионные модели



Диффузионные модели

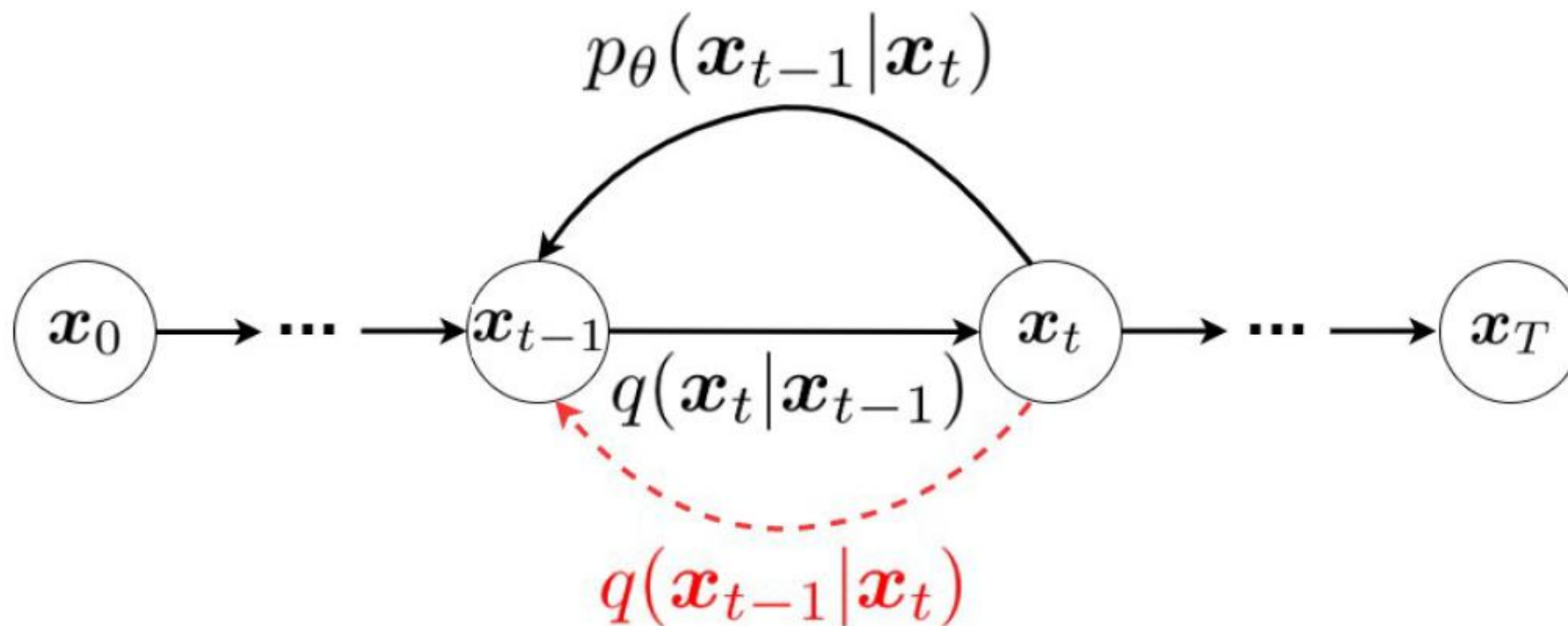


Figure 3. The conditional distributions in the forward and backward processes of the diffusion model.

Диффузионные модели

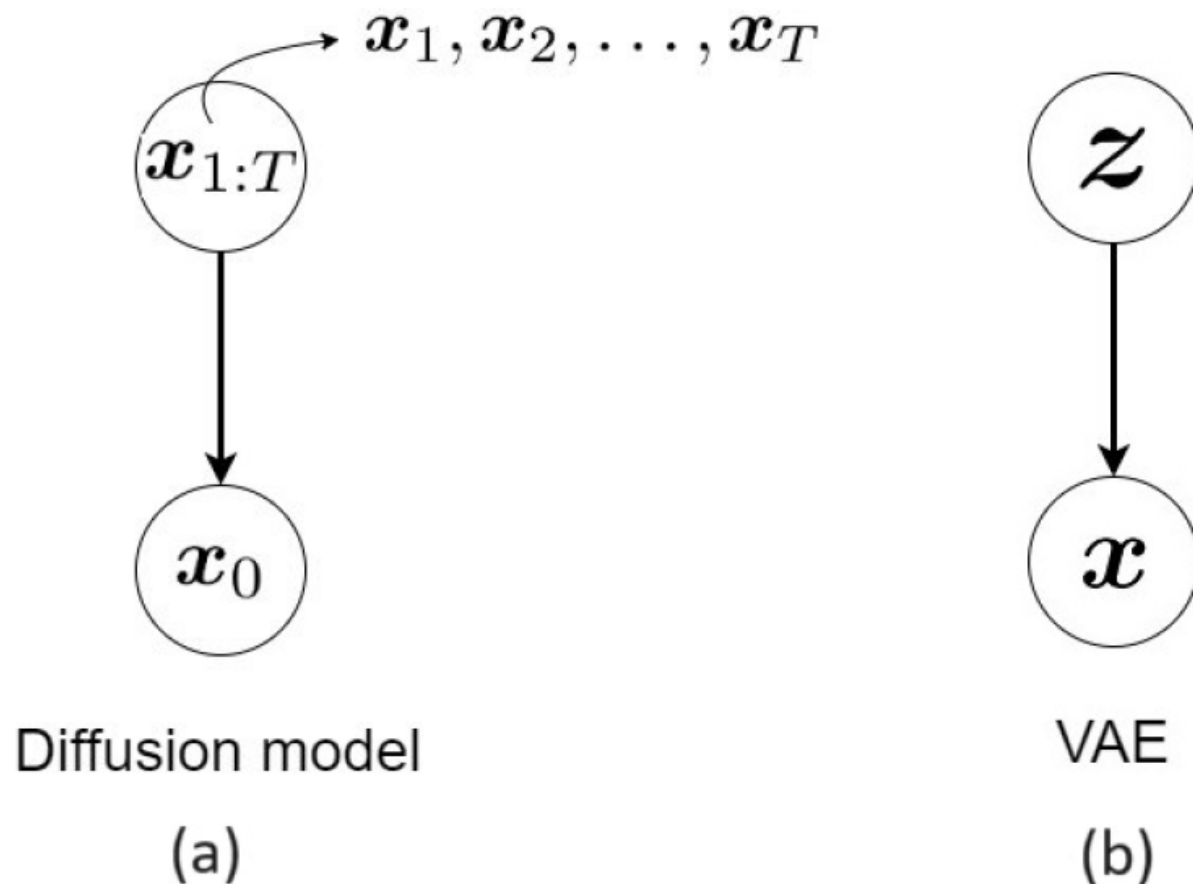
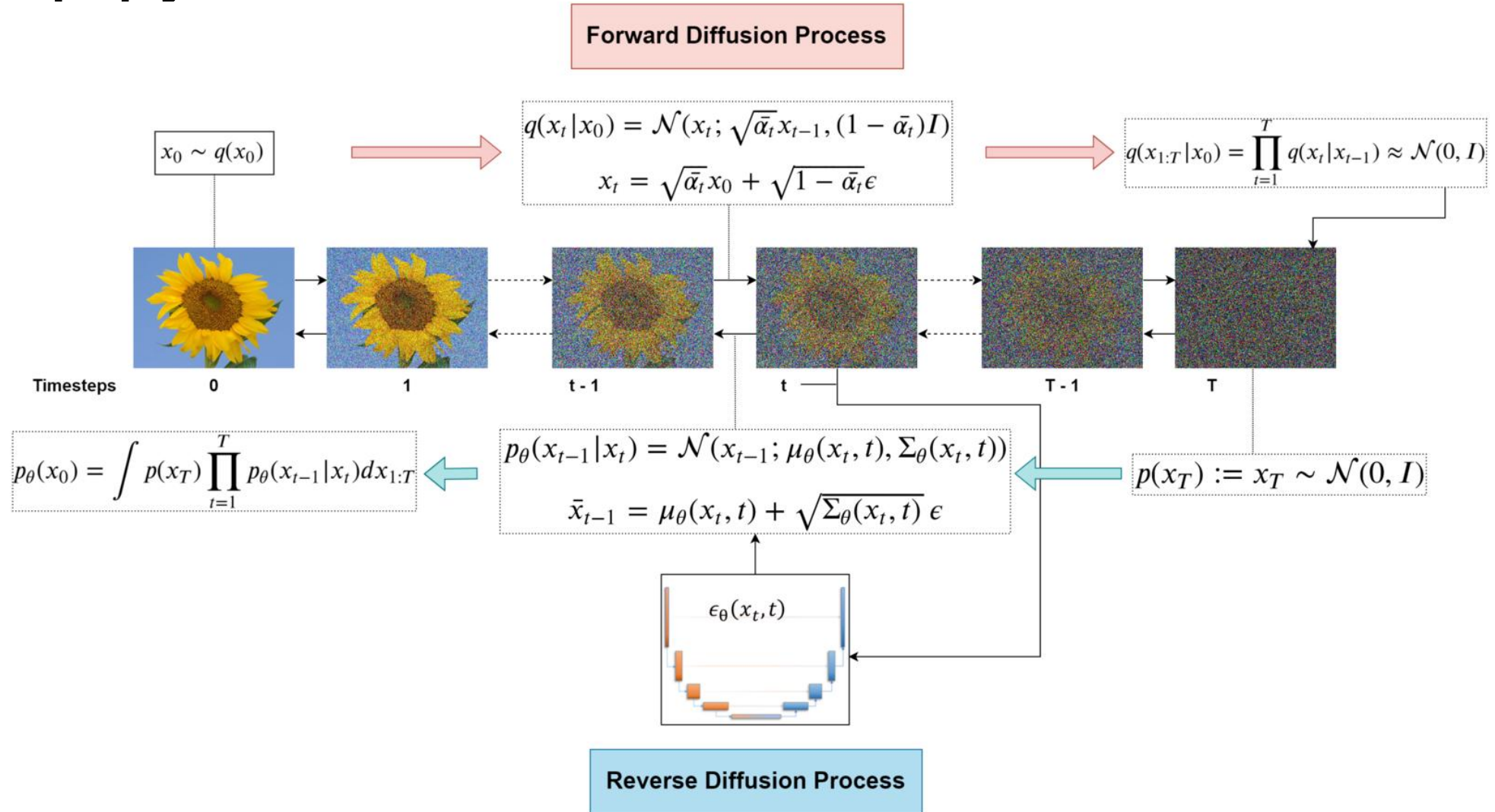


Figure 4. The probabilistic graphical models of (a) the diffusion model and (b) variational autoencoder (VAE).

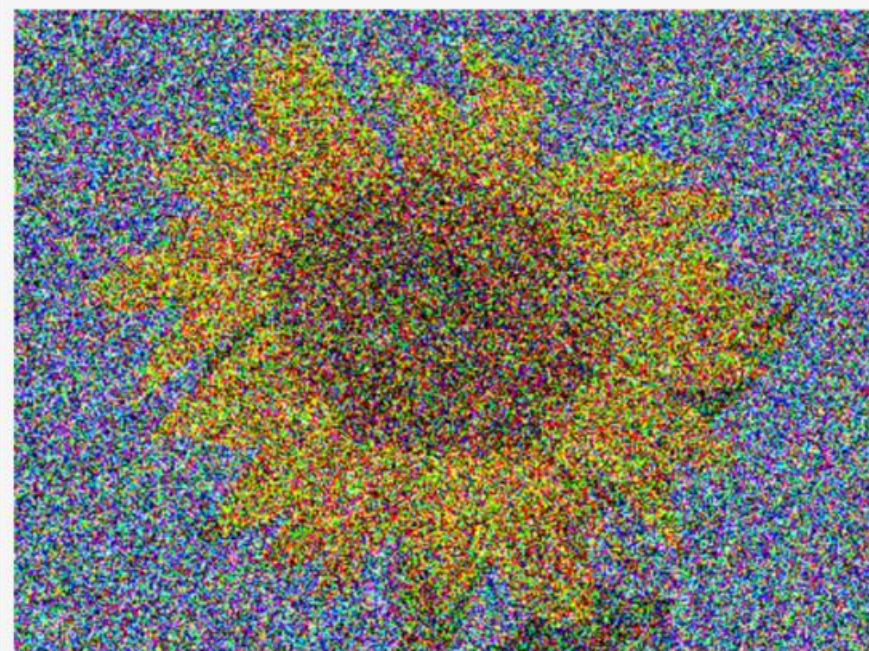
Диффузионные модели



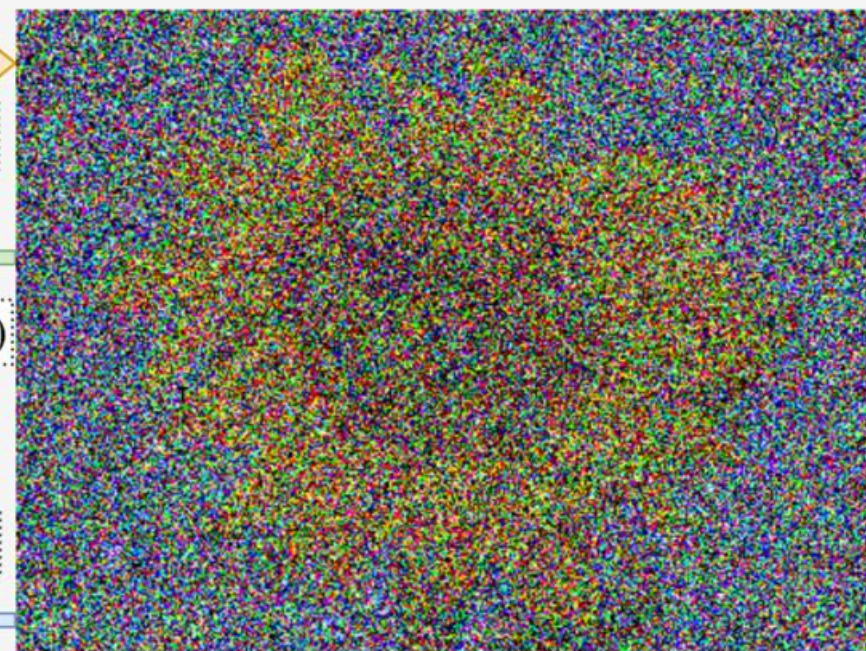
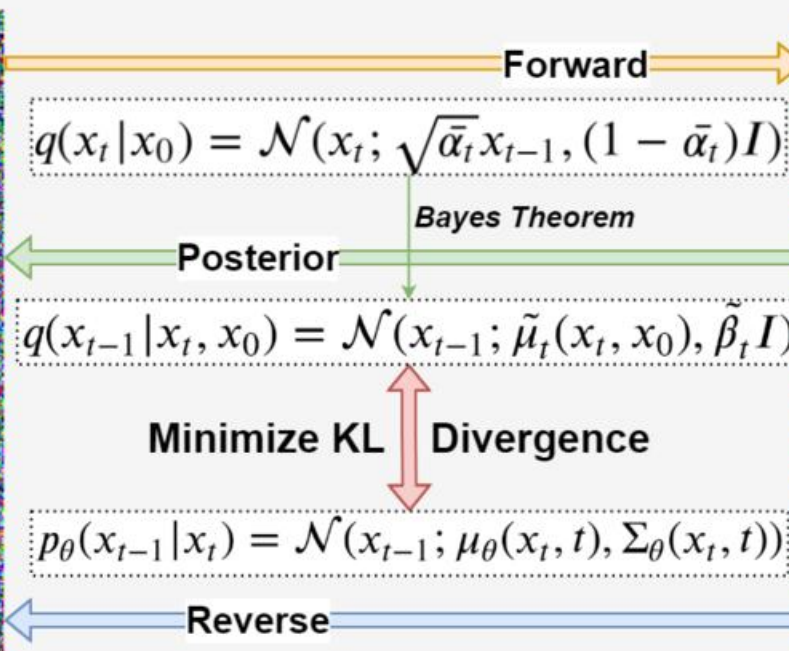
Диффузионные модели



x_0



x_{t-1}



x_t

Диффузионные модели

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\left\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

$$L_{\text{simple}} = E_{t, x_0, \epsilon} \left[\left\| \epsilon - \epsilon_{\theta}(x_t, t) \right\|^2 \right]$$

Диффузионные модели

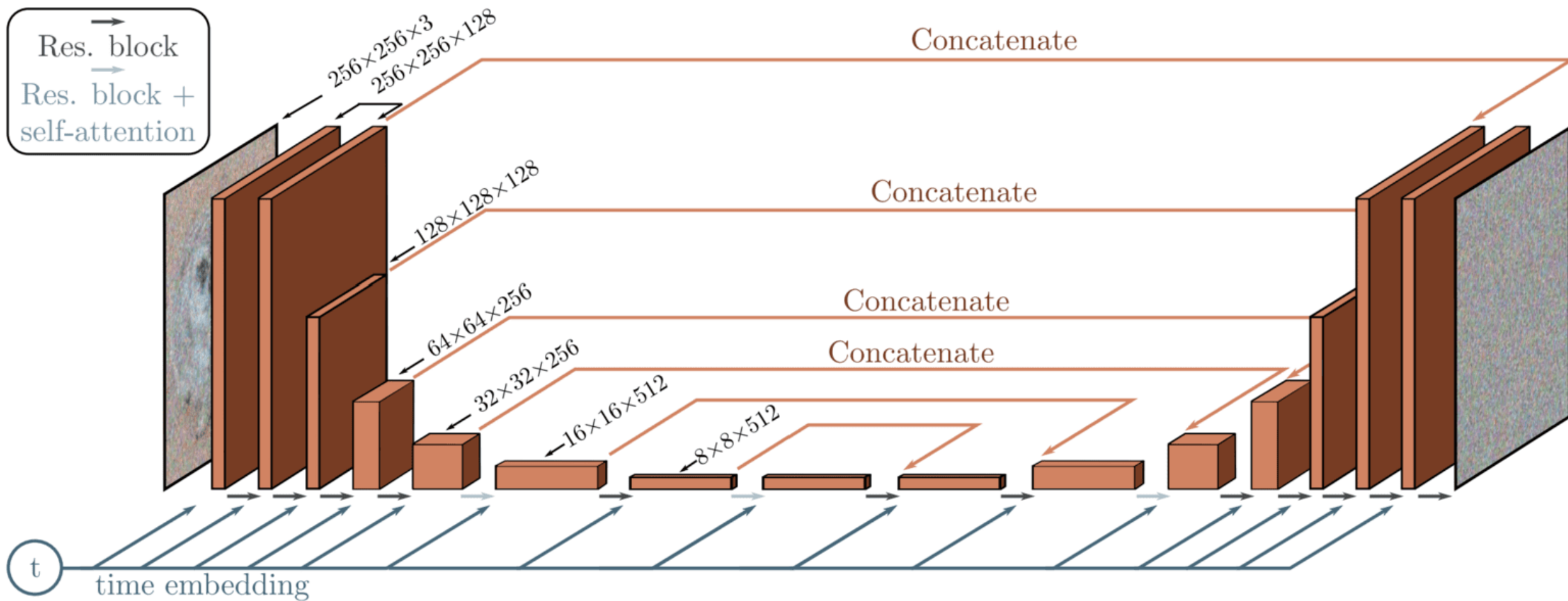
Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t) \right\|^2$   
6: until converged
```

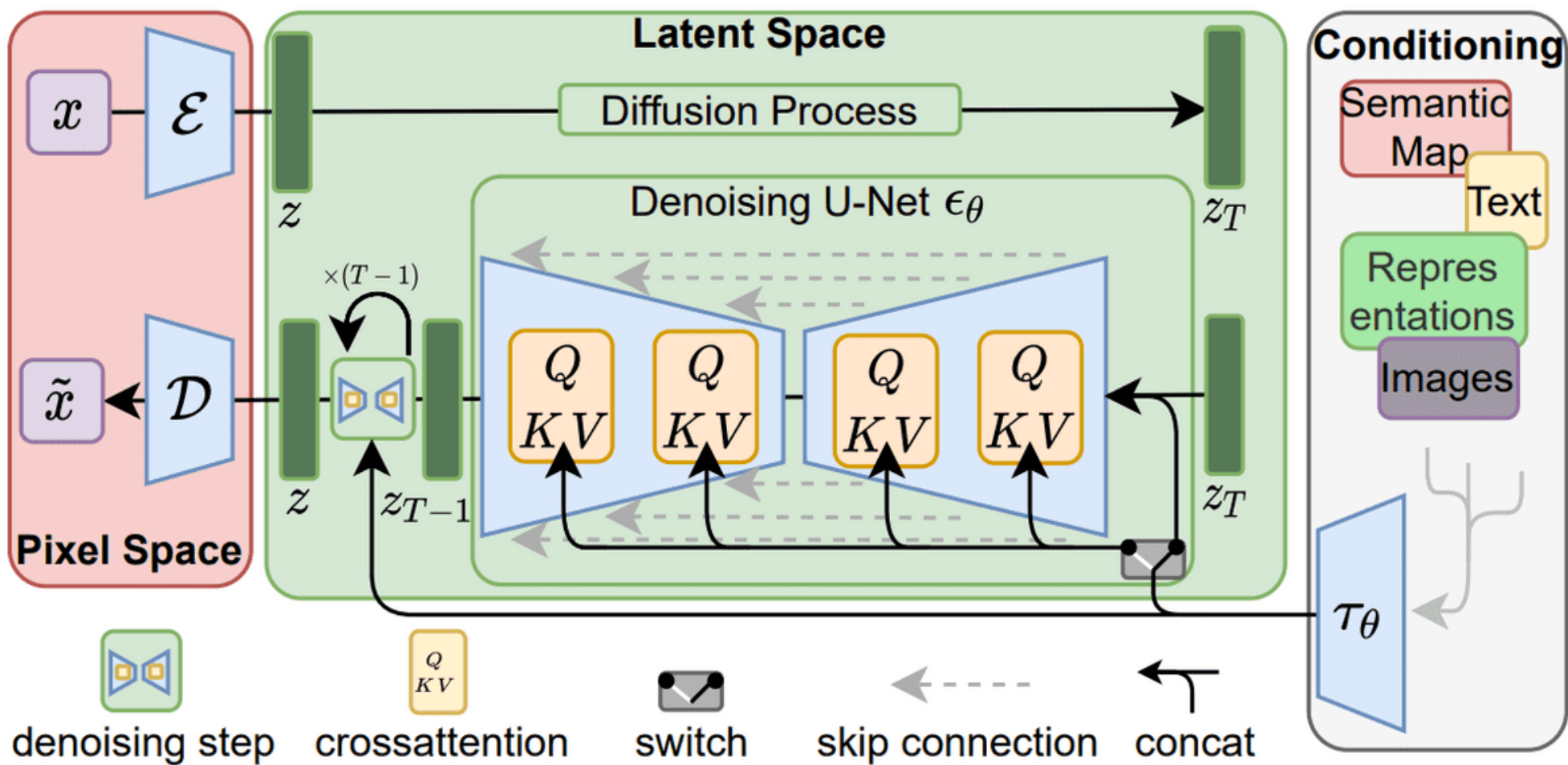
Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```

Диффузионные модели



Диффузионные модели



Диффузионные модели

[Diffusion Models: Tutorial and Survey](#)

[Lecture Notes in Probabilistic Diffusion Models](#)

[Denoising Diffusion Probabilistic Models](#)

[How diffusion models work: the math from scratch](#)

[What are Diffusion Models?](#)

[An In-Depth Guide to Denoising Diffusion Probabilistic Models DDPM – Theory to Implementation](#)

[Diffusion Course](#)